

Design and Evaluation of a Dual-Spectrum Low-Cost Sky Imager for Enhanced Solar Prediction

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Abstract

Solar forecasting is the largest source of renewable energy. However, efficiently integrating solar power into grids is difficult due to its highly intermittent nature. This can be mitigated by accurate solar forecasting. Precise short-term estimation and forecasting of solar irradiance are crucial for reliable photovoltaic (PV) and microgrid systems. To date, much research has been conducted on building solar data acquisition systems; however, most proposed systems acquire single-spectral data, such as RGB or Infrared (IR). In addition, these are generally highly expensive. In our research, we aim to design and evaluate a low-cost, dual-spectrum data acquisition (DAQ) system to capture synchronized RGB, Near Infrared (NIR), and environmental data for enhanced solar irradiance estimation. We developed a dual-camera DAQ integrating a visible-light (RGB) and a near-infrared (NIR) imaging module, each connected to a Raspberry Pi 4B unit with a BH1750 (illuminance) and an SHT40 (temperature–humidity) sensor. Both systems are housed in a custom IP56 weatherproof enclosure to ensure outdoor durability. The device records the data in a time-synchronized manner at one-minute intervals. We utilized the collected dataset to train our proposed efficient convolutional neural network (CNN) solar irradiance estimation model to assess the system’s efficacy. We trained the same

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architecture on RGB and Near Infrared (NIR) data separately, then performed post-training fusion. Among the ensemble methods, manual-ensemble achieved reduced mean squared error (MSE) by 6.8 absolute points ($\approx 12.5\%$) compared to the RGB-only baseline, and by 61.1 points ($\approx 56\%$) relative to the NIR-only model. In this research, we demonstrate a compact, affordable, and improved solar data acquisition system. The proposed DAQ system provides a foundation for the widespread adoption of improved solar monitoring technology and supports efficient renewable energy integration and localized climate analytics.

Keywords: Dual-spectrum imaging, Solar irradiance estimation, Low-cost sky imager, Multimodal data acquisition system, Infrared Imaging

1. Introduction

Solar energy stands at the forefront of national energy strategies, offering a clean, sustainable, and abundant alternative to fossil fuels. Its widespread deployment plays a pivotal role in reducing CO₂ emissions and contributing to a less carbon-intensive future, which is essential amid the ongoing global energy crisis [1]. However, the intermittent and unpredictable nature of solar power, due to daily and seasonal variations in sunlight and atmospheric conditions, poses a major challenge to its reliable integration into modern power grids [2]. To overcome this, accurate and accessible solar forecasting solutions are essential. These tools help improve the reliability of solar energy systems and facilitate their seamless incorporation into the existing energy infrastructure [3].

Solar forecasting is the process of predicting the amount of solar energy available at a specific location and time. Forecasting techniques are generally classified into three main categories: time-series statistical methods, physical models, and ensemble approaches [4]. Physical models, such as Numerical Weather Prediction (NWP), satellite imaging, and sky imagery, use atmospheric simulations and observational data to forecast solar irradiance, particularly effective for weather-based predictions [5]. Time-series statistical methods, including machine learning techniques such as Artificial Neural Networks (ANNs) and

Support Vector Machines (SVMs), leverage historical solar and meteorological data to learn patterns and relationships for predictive modelling. Ensemble methods combine both statistical and physical models to enhance forecast accuracy by leveraging the strengths of different approaches [4]. Although these approaches have improved prediction accuracy, effective solar forecasting relies on the availability of accurate, high-resolution, and relevant data [6]. Due to their ability to provide valuable information for high-quality solar forecasting, ground-based sky imagers or data acquisition systems (DAQs) are increasingly being developed [7, 8, 9, 10, 11].

Ground-based data acquisition systems (DAQs) can capture hemispherical sky images to study cloud cover, movement, and morphology. These are crucial for short-term solar irradiance forecasting [12]. While satellite imagery provides broad regional information, it suffers from low temporal frequency, limited spatial resolution, and limited precise localised information. Environmental sensors, such as pyranometers, temperature–humidity probes, and anemometers, are also used to complement accurate irradiance forecasting [13]. While these imagers/DAQs are widely used for research, their high cost limits widespread adoption. Although studies on building low-cost sky imagers have been conducted, these rely solely on RGB/NIR cameras [11, 14], omitting other spectral bands that could capture richer cloud data. High-end systems like Girasol [10] have been proposed that can capture RGB and Long-Wave Infrared (LWIR) imagery to track cloud temperature and thermal structure. However, these sensors are expensive and offer very limited spatial resolution. For instance, the FLIR Lepton 2.5 ($\approx$ \$109) captures thermal images at a resolution of 80×60 pixels [15]. On the contrary, the IMX219-160 8 MP IR-CUT camera module, featuring a 162° field of view and an integrated IR-CUT filter, provides high-resolution 8-megapixel imaging suitable for platforms such as the NVIDIA Jetson Nano and Raspberry Pi Compute Module. It offers both visible and near-infrared capture capabilities at an affordable cost of approximately \$25. While LWIR and NIR images have their pros and cons, the significantly lower cost makes NIR cameras suitable for dual-stream data acquisition while keeping the DAQ

affordable. Despite this advantage, existing literature lacks a low-cost system that combines RGB + NIR imaging. These limitations underscore the need for an affordable, time-synchronized multimodal data-acquisition system that integrates RGB + NIR sky imaging with environmental sensing to enable fine-grained, scalable, and cost-efficient solar irradiance forecasting. The research gaps can be summarized as follows,

- Existing low-cost sky imagers rely solely on either RGB or Near Infrared (NIR) sensors, while high-end systems include LWIR cameras but are prohibitively expensive and limited in resolution.
- No existing affordable platform synchronously integrates RGB and NIR imaging with environmental sensing. Hence, the existing literature lacks a scalable, time-aligned, multimodal DAQ suitable for fine-grained solar irradiance forecasting.
- There is a need to evaluate NIR’s added predictive value in solar irradiance estimation/forecasting.

In this research, we aim to address the aforementioned issues by proposing a low-cost dual-spectrum data-acquisition system for enhanced solar estimation and forecasting. The system consists of Raspberry Pi 4Bs with modular sensors (BH1750 and SHT40) and dual cameras (IMX477 and IMX219-160), housed in a custom IP56 weatherproof enclosure. It captures synchronized data with timestamp-aligned image and sensor data. We modified the NIR camera to maintain continuous, accurate cloud reflectance collection. Following this, we conducted a detailed feature analysis of the captured data. As a proof of concept, we built a lightweight solar estimation model and trained it on initially collected NIR/RGB data, followed by a post-training fusion. The key contributions of our study are as follows:

- We propose a low-cost, dual-spectrum data-acquisition system that integrates RGB and Near-Infrared (NIR) cameras with environmental sensors for synchronized solar irradiance monitoring.

- We developed a data-acquisition framework enabling minute-level, timestamped data collection with dynamic exposure control.
- We propose a lightweight convolutional neural network that employs depth-wise separable convolutions to estimate solar irradiance independently of RGB and NIR images.
- Finally, we conducted post-training fusion of the RGB and NIR models using multiple ensemble methods (mean, manual, performance-weighted, harmonic, and geometric), demonstrating improved prediction accuracy through combined spectral information.

To the best of our knowledge, no other studies have proposed a dual-spectrum sky imager that integrate RGB and Near-Infrared (NIR) with environmental sensors. This study bridges the gap between low-cost imaging platforms and high-end multispectral systems and offers a practical pathway toward scalable, real-time and remote solar monitoring. The remainder of this paper is organized as follows, Section 2 describes the related works. Section 3, 4, 5 presents the construction of the device, the software system and the solar estimation predictive modeling respectively. Lastly, Section 6 describes the experiments and results which follows the conclusion at Section 7.

2. Related Works

2.1. Ground-Based Sky Imaging and Environmental Data Acquisition

Ground-based sky imagers and data acquisition systems (DAQ) acquire hemispherical sky images at regular intervals, providing high temporal and spatial resolution for short-term solar irradiance forecasting [12]. Most ground-based sky imagers use wide-angle or hemispherical optics to capture the full sky condition. This enables the acquisition of cloud structure and irradiance pattern analysis (citeopen-source-sky-image-datasets,ground-based-sky-image). The Total Sky Imager (TSI) employs a chrome-plated hemispherical mirror to

reflect the sky into an upward-facing CCD camera. All-Sky Imagers (ASI) generally use a fisheye lens mounted within a weatherproof housing. This can typically capture the entire sky. Although fisheye optics can provide a full-sky view, it introduces optical distortions that require geometric corrections [16]. However, some sky imagers employ lenses that have a moderate field of view (FOV) ($\approx 160^\circ$) to balance coverage with reduced distortion. Apart from the distortion created by all-sky imager lenses, sun glare is another major issue in sky imaging. The Total Sky Imager (TSI) comprises a sun-tracking shadow band that blocks direct sunlight. This minimizes sun glare and ensures a consistent radiometric quality [17]. Another widely applied solution in sky imaging to mitigate saturation and recover features under high-contrast illumination conditions is the use of High Dynamic Range (HDR) [18]. HDR captures images in multiple exposures and then merges them to extend dynamic range and preserve both bright solar and dark cloud details. While optical configuration determines the spatial coverage and radiometric consistency of sky imagers, the choice of spectral band defines the nature of information captured. Consequently, RGB and NIR imaging systems have been widely employed to exploit their distinct sensitivities to atmospheric and cloud features. RGB and NIR imaging systems capture different segments of the electromagnetic spectrum, each providing unique information about the atmosphere and sky conditions. RGB cameras capture the visible spectrum ($\approx 400\text{--}700\text{ nm}$), producing red, green, and blue values that mimic human vision. These cameras are commonly used in sky imaging to identify cloud cover, cloud type, and sky clarity. The RGB values of pixels in a sky image correlate strongly with solar irradiance: blue sky pixels (e.g., $\text{RGB} \approx (0, 0, 255)$) typically indicate high irradiance, while white clouds ($\approx (255, 255, 255)$) and gray clouds ($\approx (150, 150, 150)$) are associated with reduced irradiance [19]. In contrast, NIR imaging captures wavelengths just beyond the visible spectrum, typically between 700 and 1000 nanometers. NIR imaging offers several advantages over standard RGB imaging for solar irradiance forecasting. NIR sky images significantly reduce saturation in the circumsolar region, allowing cloud data around the sun that could be washed out by the sun’s brightness

in RGB images to be preserved. Furthermore, NIR imaging enables improved detection of cloud layers and segmentation under complex sky conditions, as NIR light is less prone to atmospheric scattering and haze. NIR can also detect heat energy emitted by clouds, which can help indicate their thickness and altitude. These characteristics make NIR imaging useful, especially for extracting physical cloud features. [20]. In addition to optical and spectral imaging techniques, various environmental sensors are commonly deployed to capture meteorological parameters that influence cloud formation and solar radiation. These instruments provide complementary ground-truth data essential for accurate irradiance modeling and forecasting. Ceilometers are ground-based, active remote-sensing devices that measure the height of cloud bases using laser pulses. They provide accurate data across multiple cloud layers, even under limited visibility conditions, which is vital for understanding the impact of clouds on solar irradiance [21]. Wind speed also affects solar forecasting, as it affects cloud movement. Wind speed sensors, or anemometers, measure wind velocity, which influences cloud movement and, consequently, solar irradiance levels. Accurate wind measurements are crucial for predicting changes in cloud movements and solar energy availability [22]. Humidity sensors monitor the amount of moisture in the air, which affects cloud formation, atmospheric transmittance, and solar radiation levels. Higher humidity can increase light scattering, reducing the amount of solar energy that reaches the surface. Studies have shown that relative humidity data can be used to estimate solar irradiance with high accuracy, making it a valuable input for solar forecasting models [23]. Temperature sensors measure the ambient air temperature, which influences cloud formation and atmospheric transmittance. Changes in temperature affect the amount of solar radiation reaching Earth’s surface and are commonly used in models that estimate solar irradiance. When combined with humidity data, temperature has been shown to enable accurate predictions of hourly solar radiation [23]. For low-cost sky imagers, real-time cloud uploading via Ethernet or Wi-Fi is often ideal for remote environmental monitoring systems, enabling live data access and centralized analysis. However, in many outdoor or remote settings, net-

work connectivity may be intermittent or entirely unavailable. In such cases, local data storage becomes essential. Common approaches include writing data directly to onboard storage, such as SD cards or to external USB drives [11], which offer high-speed data access and substantial capacity for high-frequency data logging tasks. This offline-first architecture is well-suited for edge devices, enabling systems to function independently of network infrastructure. It is commonly adopted in sky-imaging and environmental-monitoring systems, such as the WAHRSIS platform, which relies on local data logging with periodic batch uploads or manual retrieval when internet access is unavailable [14]. These designs improve system reliability by eliminating reliance on real-time connectivity, reducing the risk of data loss from transmission failures.

2.1.1. Evolution of Sky Imaging Systems

Ground-based sky-imaging systems play a crucial role in solar irradiance prediction by providing cloud information. The earliest system was built by Evans and Coombe in 1959 [24, 25]. Later, much work was done on building data acquisition systems, which played an important role in solar energy utilization research. More recently, Rémi et al [7] proposed a novel sky imaging system for producing high dynamic range (HDR) sky images. They aimed to optimize the operation of Concentrated Solar Power (CSP) plants by obtaining precise short-term forecasts of solar irradiance. The system was based on a 4-megapixel color camera integrated with a fisheye lens. The key components included a neutral density filter, with faster acquisition and analysis enabled by real-time processing. In 2019, Cadlas and Alonso-Suarez [8] advanced this field by developing an all-sky imager with real-time solar irradiance measuring capabilities to predict Global Horizontal Irradiance (GHI) in short intervals. It was equipped with a DSLR camera, including an 18-megapixel CMOS sensor and a fisheye lens. The method displayed better performance than traditional models, particularly under partly cloudy conditions. The system could forecast GHI up to 10 minutes ahead. Dev. et al. [9] introduced a model based on whole-sky cameras, leveraging wide-angle, high-resolution imaging systems.

This enabled accurate short-term solar irradiance forecasting by capturing rapid fluctuations in solar radiation. The authors highlighted the benefits of the Wide Angle High-Resolution Sky Imaging System (WAHRISIS) for capturing sky images over conventional pyranometers. Recently, Terrén-Serrano et al. [10] proposed a dual-spectrum imaging for microgrids. The main novelty of the data acquisition system is that it includes an infrared camera with a visible one. For capturing sky images, long-wave infrared cameras are used, detecting radiation with a nominal response wavelength band of 8 to 14 μm . The infrared camera is a Flir Lepton 2.5 long-wave infrared camera. The cameras are mounted on a solar tracking system that keeps the sun centered in the images. This provides information on cloud dynamics. The Low Cost and Miniature ground-based Sky Camera (LAMSkyCam) [11] is designed to capture high-resolution images of the sky. The Lamsky’s key design goals were affordability and compactness; this was achieved by using 3D-printed components and store-bought electronics. It used a Raspberry Pi HQ camera with a wide-angle lens. It emphasised easy installation, achieved by making it pole-mountable. It also incorporated cooling mechanisms and local storage and backup capabilities, making it suitable for remote deployment. Overall, it is a compact, cheap, and cleverly designed system capable of capturing high-resolution images in both daylight and darkness and storing them on the device. Figure. 1 demonstrates images of the Lamsky, Girasol and our proposed Data Acquisition Systems.

2.1.2. Solar Irradiance Forecasting and Estimation

In the pursuit of precise solar potential estimation from sky images, researchers have investigated various approaches. In their study, Dev. et. al[9] incorporated cosine-weighted hemispheric sampling to simulate the pyranometer’s behavior based on a fish-eye camera lens. They calculated the relative luminance from the image and derived an empirical fitting function to scale the predicted luminance to match the irradiance values measured by the pyranometer. Huaiguang Jiang et al. [26] address GHI measurement by introducing a CNN-regression approach whereby it can be directly estimated from sky images

under various sky conditions. This offers a more efficient and potentially more widespread solution. Fei Wang, et. al [27] address the problem of rapid fluctuations in solar power generation due to different sky conditions. They propose two deep learning models (a CNN and an LSTM) to map sky images with solar irradiance data for ultra-short-term solar power forecasting (include e.g. with an example of what ultra-short-term means - e.g. $< 0.1s$). This challenge was further addressed by Song et al. [28] by implementing a real-time GHI forecasting model. The primary focus was to enhance the accuracy by incorporating a Convolutional Block Attention Mechanism (CBAM). This effectively extracts cloud features from sky images. Another interesting contribution in this field was made by Wen et al. [26] who emphasized photovoltaic power ramp-rate control (PRRC). They introduced a deep learning-based multi-step forecasting (MSF) scheme by utilizing an ensemble of convolutional neural networks. The model predicts solar power at multiple future time steps and enables effective smoothing of photovoltaic power generation. Further, Terrén-Serrano et al [10] utilizes the Girasol DAQ for a fusion of data from multiple sensors and featured sources through a multi-task deep learning architecture [29]. Building on this, Nijhum et al explore the potential of infrared images in predicting real-time solar irradiance as well as processing these type of low-res images for capturing their full potential [30, 31, 32, 33, 34]. Several research have been done on cloud cover estimation as well, which is generally done by cloud segmentation. Lightweight image segmentation methods have gained attention for enabling efficient, real-time performance in resource-constrained environments, with numerous studies proposing optimized architectures, knowledge distillation frameworks, and onboard implementations to reduce computation while maintaining accuracy [35, 36, 37, 38, 39, 40, 41, 42, 43]. Solar forecasting additionally leverages diverse environmental sensor data [44]. .



Figure 1: Comparison between existing sky imaging and data acquisition systems: (a) LAM-SkyCam [11], (b) Girasol [10], and (c) the proposed dual-spectrum DAQ system.

3. Construction of the device

In this chapter we provide an overview of the sensing and computing hardware that form the core of the environmental data acquisition device. The system integrates multiple sensors and cameras with Raspberry Pi units for synchronised data logging.

Category	Device	Purpose
Environmental Sensor	BH1750 (Lux Sensor)	Measure ambient light (lux)
Environmental Sensor	SHT40 (Temp & Humidity Sensor)	Measure temperature and humidity
Imaging	Waveshare IMX219-160 IR-Cut (NIR)	Capture near-infrared images
Imaging	Raspberry Pi HQ Camera (IMX477)	Capture visible-light (RGB) images
Data Processor	Raspberry Pi Model 4B (A)	Collect and timestamp data from NIR and lux sensor
Data Processor	Raspberry Pi Model 4B (B)	Collect and timestamp data from RGB and temp/humidity sensor
Storage	External USB Drive (2 TB)	Store NIR images and lux readings
Storage	External USB Drive (128 GB)	Store RGB images and environmental data

Table 1: Summary of hardware components used in the dual-node solar data acquisition system.

3.1. System Data Flow and Architecture Overview

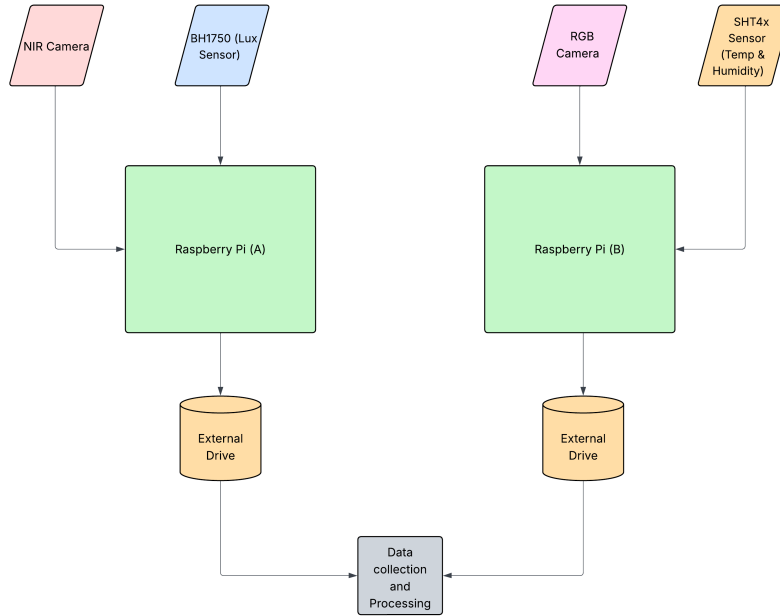


Figure 2: Data Flow Diagram for Data Acquisition Device

The overall data acquisition and processing pipeline is designed to ensure synchronized collection of environmental sensor readings and dual-spectrum (RGB–NIR) image data for reliable solar irradiance analysis. The workflow is structured as follows:

1. Sensor and Camera Input:

- The BH1750 sensor transmits ambient light intensity (lux) to Raspberry Pi A.
- The SHT40 sensor provides temperature and humidity readings to Raspberry Pi B.
- The NIR and RGB cameras capture images on Raspberry Pi A and B, respectively.

2. Processing and Timestamping:

- Each Raspberry Pi runs a script to capture data once per minute.
- All readings and images are timestamped for synchronization.
- Numeric values are saved to a .csv file as well as the path of the capture images which are captured as .jpgs.

3. Storage:

- Data is stored on external drives to enhance reliability and preserve SD card lifespan.

4. Data Aggregation and Analysis:

- Data from both drives is periodically transferred for central processing.
- Time-aligned environmental and image data is analysed to extract features and support solar irradiance prediction.

3.1.1. Dual-Pi Setup

The computational core of our proposed sky image consists of two Raspberry Pi 4 Model B (4GB RAM). Figure 2 demonstrates the dual Pi setup. One handles the NIR camera and light sensor, and the other manages the RGB camera and temperature/humidity sensor. Although a suitable adapter to connect both the cameras would be a better solution, due to the unavailability of such

a sensor, we used this setup. In future iterations, the system may be upgraded to consolidate all sensors and cameras onto a single Raspberry Pi, should appropriate hardware become available, thereby simplifying the architecture and reducing potential points of failure.

3.1.2. Environmental and Imaging Inputs.

RGB Camera. The RGB imaging device used in this system is the Raspberry Pi High Quality (HQ) Camera, which features a 12.3-megapixel Sony IMX477 sensor. The camera is connected to the Raspberry Pi via the Camera Serial Interface (CSI) and is controlled using `libcamera` software. The IMX477 sensor provides high-resolution still images with a maximum output of 4056×3040 pixels. A 6 mm 3MP HD CCTV lens is used, providing a wide field of view while maintaining sharp image quality across the frame. The lens uses a standard CS-mount and is manually focused to achieve consistent image sharpness at the target distance. Images are captured at fixed 1-minute intervals, with each capture timestamped and saved locally to an external storage device.

NIR Camera. The near-infrared (NIR) imaging system is based on the Wave-share IMX219-160 IR-Cut Camera, which utilizes an 8-megapixel Sony IMX219 sensor, allowing it to capture images with a maximum output of 3280×2465 pixels. This module originally includes an automatic IR-cut filter and side-mounted IR LEDs [45]; however, the IR lights have been removed and the cut filter disabled to allow continuous sensitivity to near-infrared wavelengths. The side-mounted IR lights provide infra-red illumination for low light conditions, but this was not needed as it is only useful for the immediate surroundings and not the sky. The cut filter allowed the camera to switch from NIR to RGB at specific light levels but this was not needed either. This keeps the camera shooting in NIR mode continuously. The camera is equipped with a wide-angle lens (160° FOV). Captured NIR images are also timestamped and stored locally at fixed intervals.

SHT40 (Temperature & Humidity Sensor). The SHT40 is a fourth-generation digital temperature and humidity sensor designed for high-accuracy, low-power environmental monitoring. It offers a relative humidity accuracy of up to $\pm 1.5\%$ RH (relative humidity) and a temperature accuracy of up to $\pm 0.1^\circ\text{C}$. The sensor can operate in a wide temperature range from -40°C to 125°C and 0% RH to 100% RH, though it performs best within the recommended range of 5°C to 60°C and 20% RH to 80% RH. Communication is handled via the I²C interface, and the sensor includes CRC checksums for data integrity. The SHT40 operates on a supply voltage of 1.08 V to 3.6 V . It features an internal heater with three power levels, enabling operation in high-humidity or condensing environments [46]. The particular sensor used also has a waterproof housing, making it suitable for deployment where rain is present incessantly. Female header pins also had to be attached to the sensor so it could connect to the pi.

BH1750 (Light Intensity Sensor). The BH1750 is a digital ambient light sensor designed for direct lux measurement with a 16-bit output. The sensor can detect illuminance in a wide dynamic range from 1 to 65,535 lux. It operates over a supply voltage range of 2.4 V to 3.6 V and communicates via the I²C interface. The sensor has three possible measurement modes: high-resolution, which can measure changes as small as 1 lux; high-resolution 2, which can measure changes as small as 0.5 lux; and low-resolution, which can measure changes as small as 4 lux. The I²C slave address can be configured as either $0x23$ or $0x5C$, depending on the state of the ADDR pin [47]. The sensor also arrived with the male header pins not attached, so solder was put in place to fasten it.

3.2. Processing and Storage

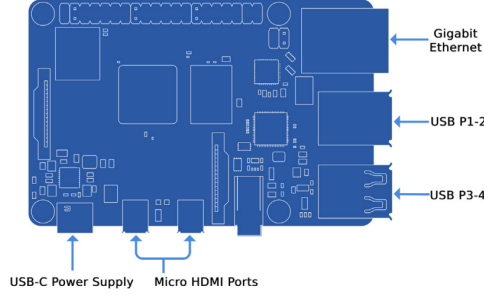


Figure 3: Raspberry Pi 4 Model B hardware layout

3.2.1. Raspberry Pi 4 Model B

The core of the system is two Raspberry Pi 4 Model B with 4 GB of RAM (see Figure 3), running Raspberry Pi OS (64-bit). The operating system was installed on a 64 GB microSD card using the official Raspberry Pi Imager tool, which enables straightforward flashing of OS images and initial configuration. The Pi 4B features a quad-core ARM Cortex-A72 processor clocked at 1.5 GHz, dual-band Wi-Fi, Bluetooth 5.0, Gigabit Ethernet, two USB 3.0 ports, and dual micro-HDMI outputs. It is responsible for sensor interfacing, image capture, data timestamping, and writing to external storage [48]. As mentioned in Table 1, Pi (A) handles NIR image capture and light sensing (BH1750) while Pi B collects RGB images and temperature/humidity data (SHT40). Each Pi independently timestamps and stores data locally.

3.2.2. External Hard Drives

We used an external USB storage device to collect the data from the Raspberry Pi. This stores environmental sensor readings and image data at an interval of 1 minute. Each data point consists of a single line of comma-separated values (CSV) and two high-resolution JPEG images—one from the RGB camera and one from the NIR camera. Each minute, the system writes approximately 2.5–5 MB of data, depending on image compression and environmental condi-

tions. Over 24 hours, this totals roughly 3.5–7 GB of data, so the current setup should be valid for at least 18 days.

3.2.3. Mechanical Design and Weatherproof Enclosure

We built a custom weatherproof enclosure to protect the system during outdoor operation. This ensures it can run safely in rain, dust, and changing temperatures. We chose a sturdy IP56-rated PVC junction box for its durability and resistance to water and dust. We further modified the lid with circular openings for mounting the RGB and NIR cameras and securely attached transparent acrylic domes, thereby providing clear optical access while keeping the interior sealed. The chosen domes are clear, lightweight, and UV-resistant. To help airflow and reduce heat buildup, we added ventilation holes, while to safely route sensor and power cables without letting water in, we used IP68 cable glands. To keep the electronics securely in place, we used a simple wooden frame and raised them slightly to prevent moisture buildup. A CAD model of the final layout was developed to assist with component positioning and dome placement and is displayed in Figure. 5. During field tests under rain and changing temperatures, it was confirmed that the design remained dry and stable, offering reliable protection and clear imaging for long-term outdoor use. Figure. 4 illustrates an overview of the enclosure displaying the assembly and modifications.

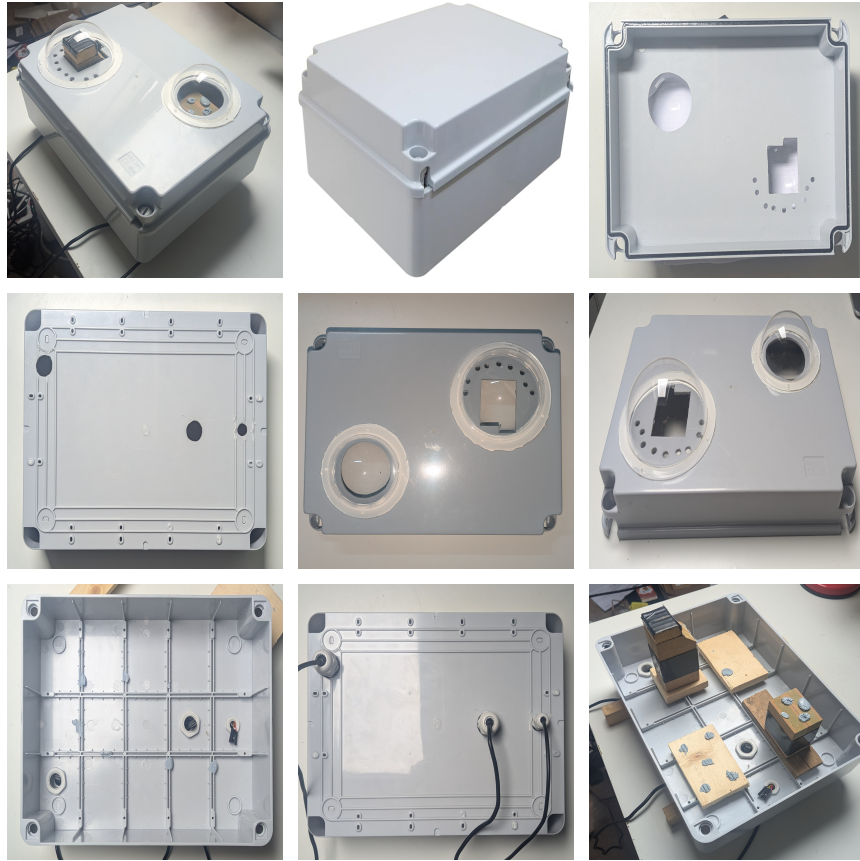


Figure 4: Overview of the custom weatherproof enclosure showing the full assembly, modifications, and internal structure.

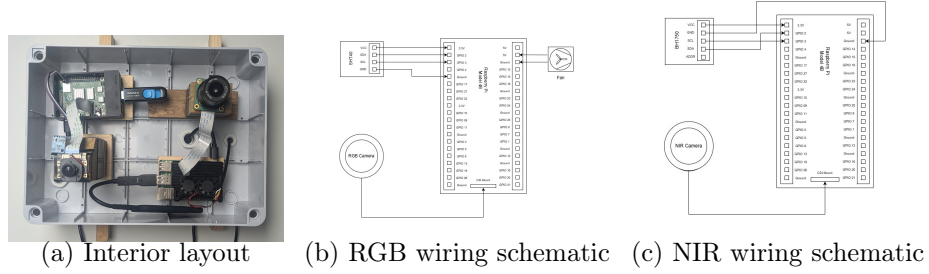


Figure 6: Illustration of the system’s physical and electrical architecture. (a) Internal layout showing Raspberry Pi units and cable routing. (b) RGB system schematic with SHT40 sensor, camera, and fan connections. (c) NIR system schematic with BH1750 sensor and NIR camera wiring.

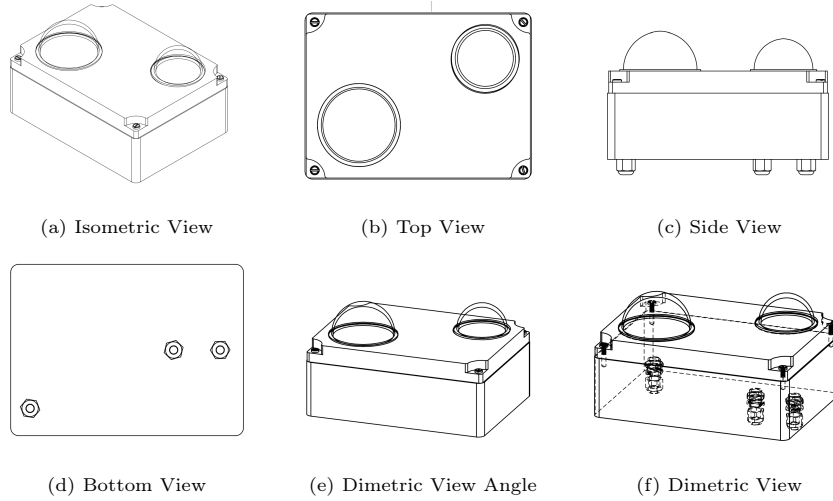


Figure 5: CAD model of the custom enclosure showing multiple perspectives: isometric, orthographic, and dimetric angles, including a seethrough rendering for internal visualization. All views are uniformly scaled and arranged in two rows.

The junction box contains two Raspberry Pi boards, mounted on custom wooden risers for airflow and protection from condensation. The internal design is separated into two units. The NIR imaging and light sensing unit consists of the NIR Camera, Raspberry Pi Model 4B (A), and the BH1750 lux sensor. We connected the BH1750 lux sensor via the I²C bus. In contrast, the

RGB imaging unit supports the RGB Camera, a Raspberry Pi Model 4B (A), and an SHT40 temperature–humidity sensor, with a **5V fan** for ventilation. We connected dedicated USB SSDs to both Pis for reliable, high-speed storage and powered them via independent 5V USB-C lines routed through IP68 cable glands to maintain waterproof integrity. Remote control and data logging were managed via Wi-Fi SSH access. This setup provided a robust, modular, and self-sustaining platform for autonomous environmental data collection in varied outdoor conditions. The real internal layout of the connections, the wiring for the Raspberry Pi that captures RGB images and the wiring for the Raspberry Pi that captures NIR images are presented in the Figure. 6.

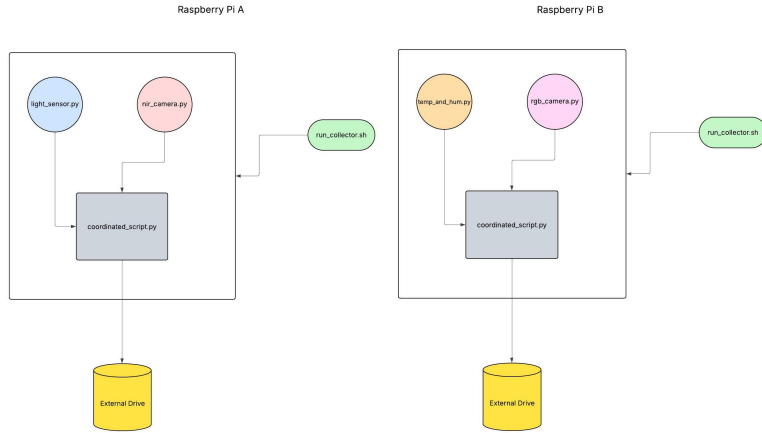


Figure 7: Software flow diagram for NIR and RGB data acquisition pipelines.

4. Software and Data Handling

We developed the proposed sky imager’s data-acquisition software to coordinate multimodal data collection across the two Raspberry Pis. The software is entirely implemented in Python for continuous outdoor operation and long-term autonomy. Each Pi collects and processes data independently yet adheres to a shared synchronized schedule, while maintaining precise temporal alignment between all sensing modalities. A flow diagram for the software is demonstrated in Figure. 7.

4.0.1. Data Acquisition and Sensor Synchronization

All data are captured synchronously from the imaging and environmental sensors at a 1-minute interval. We enabled dynamic exposure settings for the NIR camera, which automatically sets the exposure based on real-time readings from the BH1750 light sensor. This ensures consistent image quality across lighting conditions ranging from bright midday sunlight to low-light evening scenes. In contrast, automatic exposure control is set for the RGB camera. This setting performed reliably well under most daylight scenarios. In addition to the imaging sensors, the SHT40 sensor records temperature and humidity data during each capture cycle and stores them. To calculate solar irradiance, we convert the BH1750 lux readings to estimated solar irradiance using the empirical relationship proposed in [49].

$$I_{\text{solar}} = \frac{L_{\text{lux}}}{120} \quad (1)$$

This conversion allows low-cost estimation of irradiance trends without a dedicated pyranometer. The software stores each RGB and NIR image as a timestamped file. All corresponding sensor readings are automatically logged in a unified CSV file. The software verifies SSD mounts before each logging cycle and implements recovery routines for hardware or connection faults.

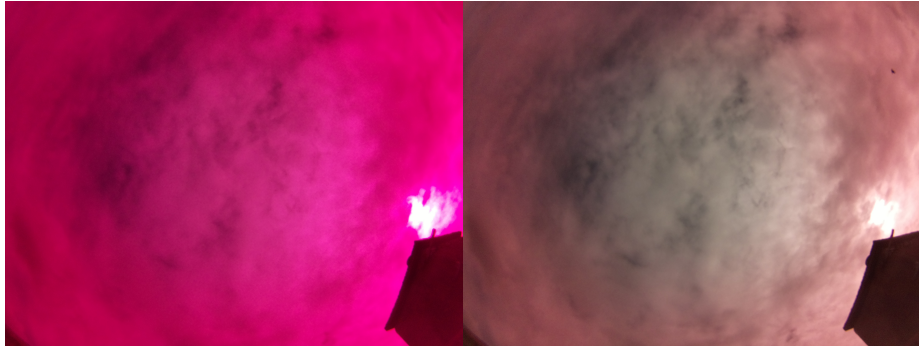


Figure 8: Effect of automatic color correction (AWB) in the camera interface. The underlying script also adjusts exposure dynamically based on ambient light, enabling consistent brightness across conditions.

5. Dual-Spectrum Solar Irradiance Estimation

One of the core contributions of this work is the inclusion of a dual-spectrum imaging system. While most of the sky imagers consist of RGB cameras only or high-end RGB and Long Wave Infrared cameras, this imager comprises a RGB camera and a Near Infrared Camera. However, it is important to test the efficacy and importance of such a system, and to determine whether using NIR images helps improve solar forecasting. Hence, we conducted a solar irradiance estimation experiment using the data collected during the initial field deployment. Since the system is in the prototype stage and the dataset is limited in size, we focused on solar irradiance estimation from sky images. The key goal was to determine whether employing Near-Infrared (NIR) image features provides additional predictive value for irradiance estimation beyond RGB imagery alone. Hence, we propose a lightweight Solar Prediction Model. In this experiment, we followed a post-training ensemble method to utilize both modalities. Two versions of the architecture were trained independently on RGB and NIR images. We trained the models to predict the paired solar irradiance values for each timestamp collected from the BH1750 sensor. We performed 4 post-training ensemble techniques: Simple Mean Blend, Manual Blend, Performance-Weighted Blend, Harmonic Mean Blend, and Geometric Mean Blend. These fusion experiments aimed to provide preliminary validation of the dual-spectrum acquisition concept.

5.1. Model Architecture.

We developed a lightweight architecture (Figure. 9) to predict the solar irradiance from the sky images. The model utilizes depthwise separable convolution, which decouples spatial and channel-wise learning while retaining high representational capacity. The network consists of 3 DepthwiseSeparableConv blocks, each of which employs a residual connection, which stabilizes training and maintains gradient flow. Each DepthwiseSeparableConv block performs a 3×3 depthwise convolution to extract spatial features within each channel.

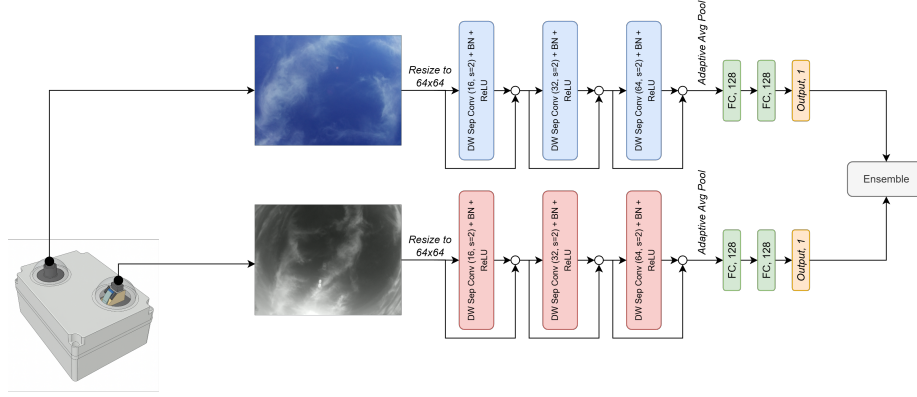


Figure 9: Architecture of the proposed SolarPredictionModel.

This follows a 1×1 pointwise convolution for channel fusion, with Batch Normalization and ReLU activation applied for a stable nonlinear representation. The output of the last DepthwiseSeparableConv block is compressed to a single latent vector by an adaptive average pooling. Lastly, this vector is passed to two multilayer perceptrons (MLP) with 128 hidden units and ReLU activation. The architecture balances accuracy and efficiency, making it ideal for real-time solar forecasting; hence, in the future, this backbone can also be integrated into forecasting models.

5.2. Training Configuration

As described in the previous section, we trained the RGB and NIR models independently. We used the Adam optimizer for training, with an initial learning rate of 1×10^{-3} . The learning rate was reduced by a factor of 0.8 every 5 epochs with the StepLR scheduler and the model was trained for 300 epochs. To minimize the difference between the predicted output and solar irradiance values, we use the mean squared error (MSE) loss. We trained on 80% of the total data and tested on the remaining 20%. We conducted all experiments using the PyTorch framework on an AMD Ryzen 7 5800X 8-core processor and an NVIDIA RTX 3060 GPU. For improved computational efficiency, Mixed-precision training was enabled via PyTorch’s automatic mixed precision (AMP). The model with the

lowest validation loss was selected and saved for final evaluation and ensemble fusion.

5.3. Ensemble Fusion Methods

In our experiment, we employed several post-training ensemble techniques. The used methods are described as follows.

5.3.1. Simple Mean Blend

Simple mean-blend averages the predictions from the NIR and RGB models for each image at each timestamp. Hence, both the model’s predictions are given equal weight in this case.

$$\hat{y}_{\text{mean}} = \frac{\hat{y}_{\text{NIR}} + \hat{y}_{\text{RGB}}}{2} \quad (2)$$

5.3.2. Manual Weighted Blend

Although the outputs can be simply adjusted by averaging. However, as the data come from two completely different modalities, where RGB images have 3 channels and NIR images have a single channel, it’s important to manually adjust the relative importance of each model’s output.

$$\hat{y}_{\text{blend}} = \alpha \hat{y}_{\text{NIR}} + (1 - \alpha) \hat{y}_{\text{RGB}} \quad (3)$$

5.3.3. Performance-Weighted Blend

The performance-weighted blend prioritizes the model with the lower validation error, assigning it a higher weight. Thus, providing a data-driven weighting without manual tuning.

$$\hat{y}_{\text{perf}} = \frac{\frac{1}{\text{MSE}_{\text{NIR}}} \hat{y}_{\text{NIR}} + \frac{1}{\text{MSE}_{\text{RGB}}} \hat{y}_{\text{RGB}}}{\frac{1}{\text{MSE}_{\text{NIR}}} + \frac{1}{\text{MSE}_{\text{RGB}}}} \quad (4)$$

5.3.4. Harmonic Mean Blend

The Harmonic Mean Blend emphasizes agreement between the two models’ predictions. It reduces the impact when a single model makes a big mistake, making it more stable and less affected by outliers or overly confident predictions.

$$\hat{y}_{\text{harm}} = \frac{2 \hat{y}_{\text{NIR}} \hat{y}_{\text{RGB}}}{\hat{y}_{\text{NIR}} + \hat{y}_{\text{RGB}} + \varepsilon} \quad (5)$$

5.3.5. Geometric Mean Blend

Emphasizing their multiplicative agreement, the Geometric Mean Blend takes the geometric average of predictions

$$\hat{y}_{\text{geo}} = \sqrt{\max(0, \hat{y}_{\text{NIR}}) \max(0, \hat{y}_{\text{RGB}})} \quad (6)$$

6. Results

6.1. Deployment Summary and Data Context

To validate and assess its performance, we deployed the data acquisition system in Longford, Ireland. The system records synchronized measurements from the attached sensors and sky cameras at 1-minute intervals. The native resolutions of the RGB and NIR cameras are 4056×3040 and 3280×2464 , respectively. We collected a total of 4960 paired clean image samples and corresponding sensor readings over 4 days. However, due to the limited time available for data collection, the current dataset is constrained by meteorological variability and by the range of cloud formations captured. Nevertheless, it serves as a proof-of-concept demonstration of the proposed dual-spectrum acquisition system. We are planning to capture longer-term, multi-seasonal datasets to improve the generalization of subsequent irradiance prediction models. Sample images of different cloud types collected from the dataset is presented in Figure. 10.

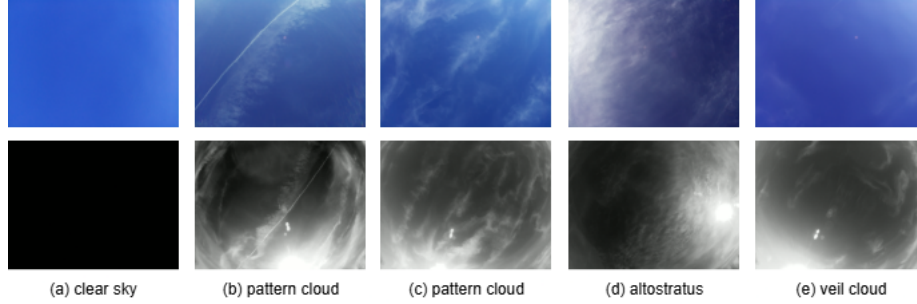


Figure 10: Sample Images collected with the RGB and Near-Infrared (NIR) cameras. The top row images are capture with the RGB while the bottom ones are acquired with the NIR camera.

6.2. Image and Sensor Feature Analysis

To evaluate the relationships among the image-derived features, environmental sensor readings, and corresponding solar irradiance. We performed a correlation analysis of the features. The resulting correlation heatmap (Fig. 11) demonstrate the correlation matrix. This illustrates how NIR, RGB, temperature, and humidity correlate with measured solar irradiance, highlighting which factors most strongly contribute to irradiance prediction. Details about each type of features are described as follows,

NIR-Derived Features.

- **NIR Brightness:** Mean grayscale intensity of the NIR image. This represents average infrared reflectivity.
- **NIR Brightness Standard Deviation:** The spread of grayscale pixel intensities, indicating contrast in the infrared spectrum.
- **NIR Red Channel Mean:** Mean intensity of the red channel, which carries the NIR signal after IR-cut lens removal.

RGB-Derived Features.

- **RGB Brightness:** Mean grayscale intensity of the RGB image, reflecting visible light brightness.

- **RGB Brightness Standard Deviation:** Standard deviation of grayscale pixel values, representing visible contrast variations.
- **RGB Red, Green, and Blue Channel Means:** Mean intensities of each channel, describing color distribution and illumination balance in visible light.

Environmental Sensor Features.

- **Temperature:** Ambient air temperature recorded by the SHT40 sensor, affecting both cloud formation and atmospheric transparency.
- **Humidity:** Relative humidity percentage from the SHT40 sensor, influencing sky opacity and irradiance diffusion.

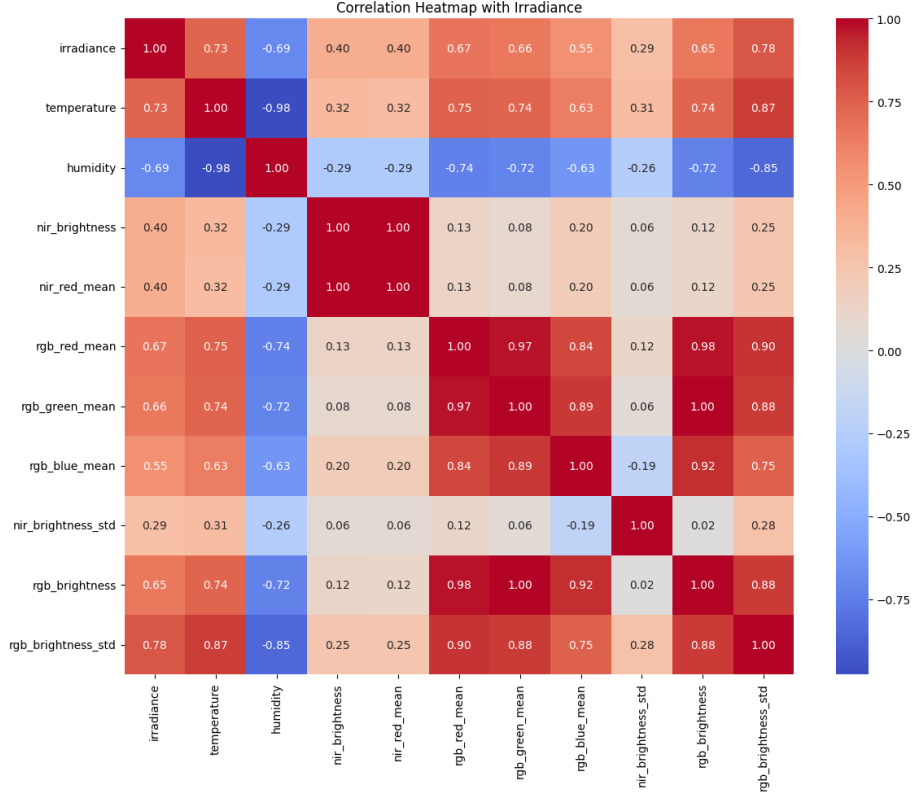


Figure 11: Pearson correlation matrix showing relationships between image-derived and environmental features with solar irradiance.

Several key patterns emerge from the correlation heatmap. **Temperature** shows strong positive correlations with irradiance. This is consistent with the physical expectation air temperature rise under high solar irradiance. **RGB-derived features** performed well in predicting irradiance, even under persistently clear, blue skies. This effectiveness stems from the RGB camera’s ability to capture subtle variations in sky color and brightness, which are influenced by the sun’s position and atmospheric interactions throughout the day. I expect this could be the camera picking up on Rayleigh scattering, where shorter wavelengths of light (blue) scatter more than longer wavelengths (red) as sunlight passes through the atmosphere. This scattering causes the sky to appear blue during midday and shifts toward warmer hues (reds and oranges) at sunrise and

sunset when the sun’s rays travel through more atmosphere. These colour shifts, even without cloud cover, are captured by the RGB sensor as variations in red, green, and blue channel intensities. For example, the red channel mean increases during sunrise and sunset due to the reddish tint in the sky, while brightness and colour balance fluctuate throughout the day with solar elevation. This makes RGB-derived features highly sensitive to the irradiance cycle, providing strong predictive power despite the lack of clouds. RGB red mean ($r = 0.67$) and RGB brightness standard deviation ($r = 0.78$) are among the strongest predictors. **Humidity** exhibits a moderate negative correlation with irradiance. This aligns with typical atmospheric behavior, where higher humidity levels coincide with reduced incoming solar radiation. In contrast, **NIR-derived features** such as `nir_brightness` and `nir_red_mean` exhibit lower correlations with irradiance ($r \approx 0.40$), suggesting limited predictive power in the current dataset.

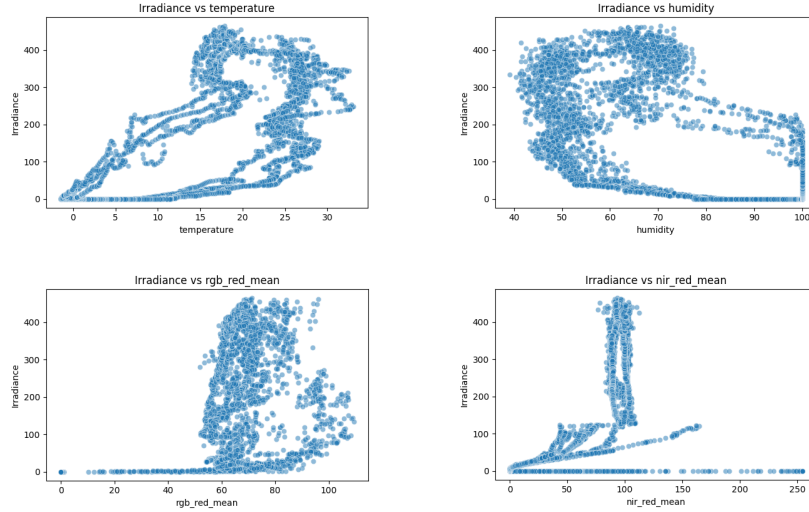


Figure 12: Scatter plots showing the relationships between solar irradiance and selected environmental and image-derived features.

The scatter plots in Figure 12 illustrate the relationships between solar irradiance and each of the collected features. Notable observations include:

- **Temperature** displays a positive relationship with irradiance, consistent

with the expectation that clear, sunny days yield both higher irradiance and temperatures.

- **Humidity** is inversely correlated, with higher humidity typically aligning with lower irradiance, likely due to cloud formation or atmospheric moisture.
- **RGB image features**, especially the red channel mean, show strong clustering with irradiance, validating their role as proxies for ambient light.
- **NIR features** provide useful variance under low-light conditions, complementing the RGB-derived data.

These observations highlight the usefulness of both sensor and image-derived features contributing to meaningful information for irradiance modeling and serve as justification for their inclusion in the predictive model.

To analyze cloud characteristics for solar forecasting, both RGB and NIR images were captured and processed. Although the system capture both RGB and NIR images they are not pixel to pixel paired images. However, the NIR camera is capable of capturing both RGB and NIR images itself, as the IR-Cut lens can be moved to change the image type. Hence to analyze the same sky condition with zero spatial and minimal temporal drift, we performed a histogram and cloud analysis. Figure 13 shows a side-by-side comparison of an RGB image and its corresponding NIR image both captured with the - Waveshare IMX219-160 IR-Cut (NIR) camera.

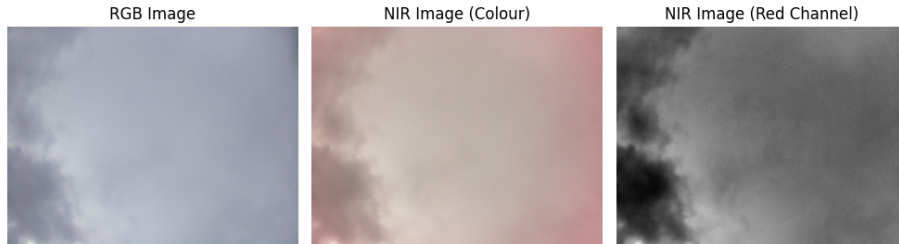


Figure 13: Left: RGB Image. Middle: NIR (All Colour). Right: NIR Image (Red Channel)

The pixel intensity distributions for the red, green, and blue channels of the RGB image, along with the NIR red channel, are shown in Figure 14. The x-axis represents pixel intensity values (0–255), and the y-axis represents the frequency of pixels with those intensities.

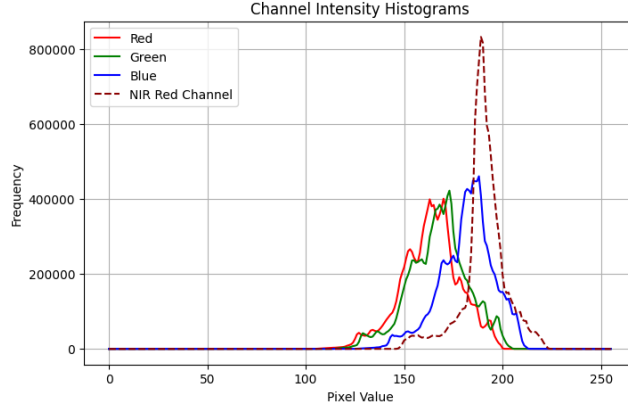


Figure 14: Channel intensity histograms for RGB and NIR images.

The NIR red channel histogram is narrower and peaks at a higher intensity (190–200) compared to the RGB channels, which peak around 160–180. This indicates that clouds reflect significantly more infrared radiation than visible light. The higher NIR intensity suggests thicker or higher-altitude clouds, which are more reflective in the infrared spectrum due to greater water or ice content. This characteristic makes NIR imaging particularly useful for identifying cloud coverage and type and this graph shows the drastic difference between the data that can be ascertained from NIR images compared to RGB.

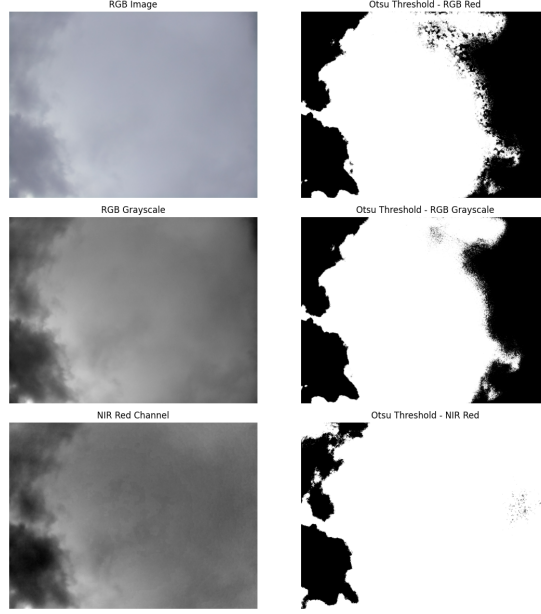


Figure 15: Cloud segmentation comparison using Otsu thresholding applied to RGB and NIR red channels.

We further performed Otsu’s thresholding to separate cloud regions from clear sky, as shown in Figure 15. The NIR red channel achieved cleaner segmentation than the RGB counterparts, confirming its superior cloud–sky separability due to differential reflectance.

The collective findings demonstrate that both image-based and environmental variables contribute complementary information to solar irradiance modeling. RGB-derived brightness and channel features capture gradual irradiance variation across clear-sky cycles, while NIR-based measures become particularly useful under diffuse or cloudy illumination. These results motivated the inclusion of both feature types in the predictive model described below.

6.3. Predictive Model Performance

As outlined in Section 5, one of the key objective of the data acquisition system is to utilize features from both near-infrared (NIR) and red-green-blue (RGB) images. In this study we introduce a lightweight Solar Prediction Model

that estimates solar irradiance directly from NIR or RGB images. In this experiment we aim evaluate the significance of dual-spectrum information fusion in solar forecasting. This section presents the evaluation and comparison of various ensemble strategies (Mean, Manual, Performance-weighted, Harmonic, Geometric) for solar irradiance prediction using outputs from NIR and RGB models.

6.3.1. Evaluation Metrics

The model's performance was evaluated using two common regression metrics: Mean Squared Error (MSE) and Root Mean Squared Error (RMSE).

Mean Squared Error measures the average squared difference between predicted and actual values:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Root Mean Squared Error is the square root of MSE:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Normalized Root Mean Squared Error (NRMSE) is the normalized form of RMSE, which expresses the error relative to the data range:

$$\text{NRMSE} = \frac{\text{RMSE}}{y_{\max} - y_{\min}} = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}}{y_{\max} - y_{\min}}$$

where:

- y_i is the actual value
- $\hat{y}_i$ is the predicted value
- n is the total number of predictions

MSE is useful for penalizing large errors more heavily due to squaring, which is helpful when outliers are important. **RMSE**, being in the same unit as the

target variable, offers a more interpretable measure of average prediction error. **NRMSE** is the normalized form of RMSE, which expresses the error relative to the data range:

Table 2: Comparison of validation metrics across different ensemble strategies for solar irradiance prediction.

Metric	NIR	RGB	Mean	Manual ($\alpha=0.3$)	Perf.-Weighted	Harmonic	Geometric
MSE	108.6	54.3	53.9	47.5	47.9	49.8	51.3
RMSE	10.422	7.371	7.341	6.888	6.922	7.060	7.161
NRMSE	0.177	0.125	0.125	0.117	0.118	0.120	0.122

6.3.2. Experimental Results

Our detailed comparison of individual and ensemble models reveals how multimodal fusion enhances prediction accuracy for solar irradiance estimation. Table 2 provides a summary of the experimental results. NIR images alone achieve suboptimal performance due to their grayscale, single-channel nature, compared to RGB images. Nevertheless, information fusion consistently improves robustness across evaluation metrics. The Manual Ensemble technique produces the best results, achieving a root-mean-square error (RMSE) of 47.5 W/m². We kept $\alpha = 0.3$, thereby assigning greater weight to the RGB model output while retaining a substantial contribution from the NIR model. Harmonic ensembles perform comparably, reducing the loss to 47.9 W/m² and demonstrating consistent improvements over single RGB or NIR modality baselines. Simple averaging (Mean) also significantly reduces error relative to the NIR-only approach, confirming the complementary nature of the two modalities.

While manual weight ensemble achieves the lowest loss, it largely depends on the assigned weight α . We tuned the hyperparameter to determine the optimal weight that yields the best result. Fig. 16 demonstrate a line chart of the results achieved for different values of alpha. It's clear that weights assigned beyond 0.3 gradually increase the root mean square error (RMSE), whereas decreasing them does the same. This demonstrates that both NIR and RGB

provide useful information, and finding the right balance between them is crucial for more accurate predictions.

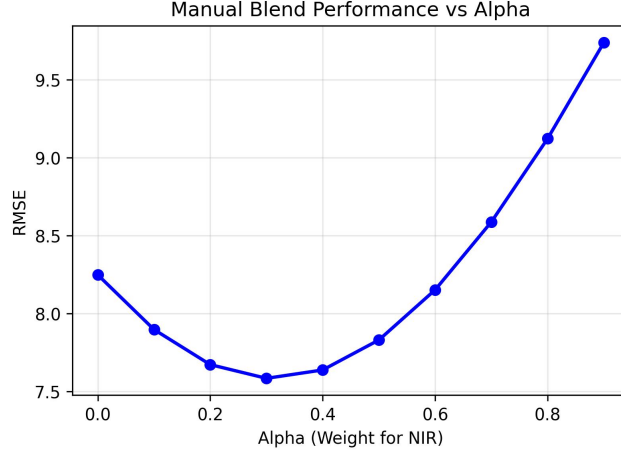


Figure 16: Performance of the manual ensemble method employing different alpha values.

6.4. Measurement Considerations

The prototype system integrates sensors with manufacturer-rated accuracies, which help maintain consistent measurement stability. The BH1750FVI ambient light sensor features a measurement variation of $\pm 20\%$ across a range of 1–65,535 lx [50], while the Sensirion SHT40 temperature–humidity module offers typical accuracies of $\pm 0.1^\circ\text{C}$ and $\pm 1.5\%$ RH [51]. Such tolerances are considered adequate for the irradiance estimation analysis conducted in this work.

7. Conclusion

In this work, we propose a dual-spectrum (RGB + NIR) data-acquisition system that integrates environmental sensors and provides synchronized multi-modal inputs for solar irradiance estimation. The sky imager is housed in a compact, weatherproof IP56 enclosure with independent Raspberry Pi nodes and modular sensor architecture for reliable outdoor operation. We built a synchronization pipeline that enables minute-level, timestamp-aligned image–sensor

data capture with dynamic NIR exposure control. To analyze the captured images, we performed a feature correlation analysis, revealing that RGB brightness and color statistics strongly correlate with irradiance, while NIR features enhance contrast and cloud detection. To assess the system’s performance, we developed a lightweight CNN regression model employing depthwise separable convolutions to estimate irradiance efficiently from both RGB and NIR images. We trained this architecture on both the collected RGB and NIR data following an ensemble fusion. This revealed that the manual blend strategy gained the best result (6.89 W/m^2), validating complementary information across modalities. This shows that the dual-spectrum method improves solar prediction. Despite limited atmospheric diversity during this short deployment, the methodology demonstrates the feasibility of autonomous, multimodal sky imaging for solar monitoring. Future deployments spanning broader temporal and weather variations will enable a richer dataset, supporting stronger model generalization and adaptive irradiance forecasting under dynamic atmospheric conditions.

References

- [1] T. S. Kishore, P. U. Kumar, V. Ippili, Review of global sustainable solar energy policies: Significance and impact, *Innovation and Green Development* (2025).
- [2] C. Wu, X.-P. Zhang, M. Sterling, Solar power generation intermittency and aggregation, *Scientific Reports* (2022).
- [3] K. Barhmi, C. Heynen, S. Golroodbari, W. van Sark, A review of solar forecasting techniques and the role of artificial intelligence, *Solar* (2024).
- [4] S. Sobri, S. Koohi-Kamali, N. Abd Rahim, Solar photovoltaic generation forecasting methods: A review, *Energy Conversion and Management* (2018).
- [5] R. H. Inman, H. T. Pedro, C. F. Coimbra, Solar forecasting methods for

- renewable energy integration, *Progress in Energy and Combustion Science* (2013).
- [6] M. Attya, O. M. Abo-Seida, H. M. Abdulkader, A. M. Mohammed, Advanced solar radiation prediction using combined satellite imagery and tabular data processing, *Scientific Reports* (2025).
 - [7] R. Chauvin, J. Nou, S. Thil, S. Grieu, Generating high dynamic range images using a sky imager, *IFAC-PapersOnLine* 50 (1) (2017) 219–224.
 - [8] M. Caldas, R. Alonso-Suárez, Very short-term solar irradiance forecast using all-sky imaging and real-time irradiance measurements, *Renewable energy* 143 (2019) 1643–1658.
 - [9] S. Dev, F. M. Savoy, Y. H. Lee, S. Winkler, Estimating solar irradiance using sky imagers, *Atmospheric Measurement Techniques* 12 (10) (2019) 5417–5429.
 - [10] G. Terrén-Serrano, A. Bashir, T. Estrada, M. Martínez-Ramón, Girasol, a sky imaging and global solar irradiance dataset, *Data in Brief* 35 (2021) 106914.
 - [11] M. Jain, V. S. Sengar, I. Gollini, M. Bertolotto, G. McArdle, S. Dev, Lamskycam: A low-cost and miniature ground-based sky camera, *HardwareX* 12 (2022). doi:10.1016/j.ohx.2022.e00346.
 - [12] S. Dev, F. M. Savoy, Y. H. Lee, S. Winkler, Estimating solar irradiance using sky imagers, *Atmospheric Measurement Techniques* 12 (2019). doi:10.5194/amt-12-5417-2019.
 - [13] K. Blazakis, N. Schetakis, P. Bonfini, K. Stavarakakis, E. Karapidakis, Y. Katsigiannis, Towards automated model selection for wind speed and solar irradiance forecasting, *Sensors* (2024).
 - [14] S. Dev, F. M. Savoy, Y. H. Lee, S. Winkler, Wahrsis: A low-cost, high-resolution whole sky imager with near-infrared capabilities, in: *Proceedings*

- of IS&T/SPIE Infrared Imaging Systems, SPIE, 2017. doi:10.1117/12.2256686.
- [15] Teledyne FLIR, Flir lepton family – thermal imaging modules, <https://oem.flir.com/developer/lepton-family/>, accessed: 2025-11-09 (2025).
 - [16] F. Lin, Y. Zhang, J. Wang, Recent advances in intra-hour solar forecasting: A review of ground-based sky image methods, *International Journal of Forecasting* 39 (2023). doi:10.1016/j.ijforecast.2021.11.002.
 - [17] Y. Nie, X. Li, Q. Paletta, M. Aragon, A. Scott, A. Brandt, Open-source sky image datasets for solar forecasting with deep learning: A comprehensive survey, *Renewable and Sustainable Energy Reviews* 189 (2024). doi:10.1016/j.rser.2023.113977.
 - [18] B. Urquhart, B. Kurtz, E. Dahlin, M. Ghonima, J. E. Shields, J. Kleissl, Development of a sky imaging system for short-term solar power forecasting, *Atmospheric Measurement Techniques* 8 (2015). doi:10.5194/amt-8-875-2015.
 - [19] F. Wang, Z. Xuan, Z. Zhen, Y. Lie, K. Li, L. Zhao, M. Shafie-khah, J. P. Catalão, A minutely solar irradiance forecasting method based on real-time sky image-irradiance mapping model, *Energy Conversion and Management* (2020).
 - [20] G. Terrén-Serrano, M. Martínez-Ramón, Deep learning for intra-hour solar forecasting with fusion of features extracted from infrared sky images, *Information Fusion* (2023).
 - [21] D. Magiera, Y. Fabel, B. Nouri, N. Blum, D. Schnaus, L. F. Zarzalejo, Advancing semantic cloud segmentation in all-sky images: A semi-supervised learning approach with ceilometer-driven weak labels, *Solar Energy* 300 (2025) 113822.

- [22] K. Blazakis, Y. Katsigiannis, G. Stavrakakis, One-day-ahead solar irradiation and windspeed forecasting with multi-channel convolutional neural networks, *Energies* (2022).
- [23] D. F. Al Riza, S. I. u. H. Gilani, M. S. Aris, Hourly solar radiation estimation using ambient temperature and relative humidity data, *International Journal of Environmental Science and Development* (2011).
- [24] F. Lin, Y. Zhang, J. Wang, Recent advances in intra-hour solar forecasting: A review of ground-based sky image methods, *International Journal of Forecasting* 39 (1) (2023) 244–265.
- [25] G. C. Evans, D. Coombe, Hemispherical and woodland canopy photography and the light climate, *Journal of Ecology* 47 (1) (1959) 103–113.
- [26] H. Wen, Y. Du, X. Chen, E. Lim, H. Wen, L. Jiang, W. Xiang, Deep learning based multistep solar forecasting for pv ramp-rate control using sky images, *IEEE Transactions on Industrial Informatics* 17 (2) (2020) 1397–1406.
- [27] F. Wang, Z. Zhang, H. Chai, Y. Yu, X. Lu, T. Wang, Y. Lin, Deep learning based irradiance mapping model for solar pv power forecasting using sky image, in: 2019 IEEE industry applications society annual meeting, IEEE, 2019, pp. 1–9.
- [28] S. Song, Z. Yang, H. Goh, Q. Huang, G. Li, A novel sky image-based solar irradiance nowcasting model with convolutional block attention mechanism, *Energy Reports* 8 (2022) 125–132.
- [29] G. Terrén-Serrano, M. Martínez-Ramón, Deep learning for intra-hour solar forecasting with fusion of features extracted from infrared sky images, *Information Fusion* 95 (2023) 42–61.
- [30] I. R. Nijhum, P. Kenny, S. Dev, Intra-hour solar irradiance estimation using infrared sky images and mobilenetv2-based cnn regression, in: 2023 IEEE

7th Conference on Energy Internet and Energy System Integration (EI2), 2023, pp. 3534–3539. doi:10.1109/EI259745.2023.10512911.

- [31] I. R. Nijhum, D. Kaloni, P. Kenny, S. Dev, Enhancing intra-hour solar irradiance estimation through knowledge distillation and infrared sky images, in: IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium, 2024, pp. 7207–7211. doi:10.1109/IGARSS53475.2024.10642288.
- [32] M. Y. Hossain, I. R. Nijhum, M. M. H. Rakib, S. Dev, Can deep learning outperform traditional methods in upscaling ir images for solar irradiance estimation?, IET Conference Proceedings 2024 (2024) 250–257. arXiv:<https://digital-library.theiet.org/doi/pdf/10.1049/icp.2024.3312>, doi:10.1049/icp.2024.3312.
URL <https://digital-library.theiet.org/doi/abs/10.1049/icp.2024.3312>
- [33] I. R. Nijhum, M. Y. Hossain, M. M. H. Rakib, S. Dev, On improving solar forecasts with enhanced ir imaging and cnns, IET Conference Proceedings 2024 (2024) 95–102. arXiv:<https://digital-library.theiet.org/doi/pdf/10.1049/icp.2024.3289>, doi:10.1049/icp.2024.3289.
URL <https://digital-library.theiet.org/doi/abs/10.1049/icp.2024.3289>
- [34] I. R. Nijhum, H. Hassan, P. Kenny, S. Dev, Multiscale deep feature fusion for solar estimation from infrared sky images, in: Proceedings of the 33rd International Conference on Artificial Intelligence and Cognitive Science (AICS 2025), Hyatt Centric Liberties Hotel, AICS, Dublin, Ireland, 2025, corresponding author: Soumyabrata Dev. Tel.: +353 1 896 1797, Office A1.16, UCD School of Computer Science, University College Dublin, Ireland.

- [35] I. R. Nijhum, M. S. Reza, S. Dev, Tiny-ucloudnet: Memory efficient ground-based cloud segmentation network, in: Proceedings of the 2025 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), IEEE, Brisbane, Australia, 2025, 3–8 August 2025. doi:TBD.
- [36] N. Beheshti, L. Johnsson, Squeeze u-net: A memory and energy efficient image segmentation network, in: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2020, pp. 1495–1504. doi:10.1109/cvprw50498.2020.00190.
- [37] H. Zunair, A. B. Hamza, Sharp u-net: Depthwise convolutional network for biomedical image segmentation, Computers in Biology and Medicine 136 (2021) 104699. doi:10.1016/j.combiomed.2021.104699.
- [38] Y. Huang, H. Yang, K. Sun, S. Zhang, L. Cao, G. Jiang, R. Ji, Interformer: Real-time interactive image segmentation, in: Proceedings of the 2023 IEEE/CVF International Conference on Computer Vision (ICCV), 2023, pp. 22244–22254. doi:10.1109/iccv51070.2023.02038.
- [39] Z. Zeng, Q. Hu, Z. Xie, B. Li, J. Zhou, Y. Xu, Small but mighty: Enhancing 3d point clouds semantic segmentation with u-next framework, International Journal of Applied Earth Observation and Geoinformation 136 (2024) 104309. doi:10.1016/j.jag.2024.104309.
- [40] D. A. Zeleke, H.-D. Kim, A new strategy of satellite autonomy with machine learning for efficient resource utilization of a standard performance cubesat, Aerospace 10 (1) (2023) 78. doi:10.3390/aerospace10010078.
- [41] K. Chatar, K. Kitamura, M. Cho, Onboard data prioritization using multi-class image segmentation for nanosatellites, Remote Sensing 16 (10) (2024) 1729. doi:10.3390/rs16101729.
- [42] Y. Zhang, Z. Chen, X. Yang, Light-m: An efficient lightweight medical image segmentation framework for resource-constrained iomt, Computers

in Biology and Medicine 170 (2024) 108088. doi:10.1016/j.combiomed.2024.108088.

- [43] J. Chen, R. Chen, W. Wang, J. Cheng, L. Zhang, L. Chen, Tinyu-net: Lighter yet better u-net with cascaded multi-receptive fields, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2024, pp. 626–635.
- [44] I. R. Nijhum, T. Akinola, M. E. Ali, S. Dev, Optimizing lightweight photovoltaic forecasting models: A distillation approach, in: Proceedings of the 2025 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), IEEE, Brisbane, Australia, 2025, 3–8 August 2025. doi:TBD.
- [45] Waveshare Electronics, IMX219-160 IR-CUT Camera – Product Wiki, https://www.waveshare.com/wiki/IMX219-160_IR-CUT_Camera, accessed: 19 April 2025 (2025).
- [46] S. AG, SHT40: 4th Generation, High-Accuracy, Ultra-Low-Power, 16-bit Relative Humidity and Temperature Sensor Platform, https://www.sensirion.com/file/datasheet/SHT4x_Datasheet.pdf, datasheet, Version 1 (October 2020).
- [47] R. Semiconductor, BH1750FVI: Digital 16-bit Serial Output Type Ambient Light Sensor IC for I2C Bus Interface, <https://www.mouser.com/datasheet/2/348/bh1750fvi-e-186247.pdf>, datasheet (November 2014).
- [48] Raspberry Pi Ltd, Raspberry Pi 4 Model B Product Brief, <https://www.raspberrypi.com/documentation/computers/raspberry-pi.html>, published by Raspberry Pi Ltd (February 2025).
- [49] P. R. Michael, D. E. Johnston, W. Moreno, A conversion guide: solar irradiance and lux illuminance, Journal of measurements in engineering 8 (4) (2020) 153–166.

- [50] ROHM Semiconductor, Kyoto, Japan, BH1750FVI Ambient Light Sensor IC for I²C Bus Interface, revision E, November 2014 (2014).
URL <https://www.mouser.com/datasheet/2/348/bh1750fvi-e-186247.pdf>
- [51] Sensirion AG, Staefa, Switzerland, SHT4x Series 4th Generation High-Accuracy Ultra-Low-Power 16-bit Relative Humidity and Temperature Sensor Platform, version 1, October 2020 (2020).
URL <https://download.mikroe.com/documents/datasheets/SHT40%20Datasheet.pdf>